SAGAR PANCHAL, PhD

Medicinal Chemistry | Chemical Biology | Protein Biophysics | Molecular Modelling

India | spanchal@chemistry.du.ac.in | ssagar61@gmail.com | +91 945-768-9739 / +91 921-175-7423

Google Scholar: https://scholar.google.com/citations?hl=en&user=RUpphiAAAAAJ | ORCID: https://orcid.org/0000-0001-5950-0729

PROFESSIONAL PROFILE

Medicinal chemistry and chemical biology researcher with experience integrating experimental biophysics and computational modelling to investigate protein–ligand interactions, protein conformational changes, and molecular recognition. Research experience includes fluorescence spectroscopy, circular dichroism, FRET, UV–Vis spectroscopy, molecular docking, molecular dynamics simulations, and MM/PBSA and MM/GBSA binding free-energy analysis.

Current research at Gujarat Biotechnology University focuses on inhibitor development against FMO3 and TMA lyases involved in atherosclerosis.

RESEARCH HIGHLIGHTS

- Integrated spectroscopic methods and molecular simulations to investigate ligand binding and conformational changes in biomolecular systems.
- Designed and synthesized noscapine-derived π-cationic ionic liquids and investigated their interactions with serum albumin and haemoglobin.
- Applied molecular docking, GROMACS molecular dynamics, protein–ligand interaction analysis, and MM/PBSA/MM/GBSA calculations.
- Experience spanning medicinal chemistry, chemical biology, protein biophysics, and cell-based cytotoxicity assays.

RESEARCH EXPERIENCE

Research Associate | *Gujarat Biotechnology University (GBU), Gandhinagar, India | 2026–Present*

- Project: “Development of Inhibitors against FMO3 and TMA Lyases Involved in the Progression of Atherosclerosis” (DBT-funded project).
- Research focus: development and evaluation of potential inhibitors of enzymes involved in the gut-liver trimethylamine/trimethylamine-N-oxide (TMA/TMAO) pathway.
- Relevant interests include protein-ligand interactions, molecular modelling, and structure-guided inhibitor discovery.

PhD Researcher, Chemistry | *University of Delhi, India | 2020–2026*

- Designed and synthesized noscapine-derived π-cationic ionic liquids for bioactive and anticancer applications.
- Integrated fluorescence spectroscopy, circular dichroism, FRET, and UV-is spectroscopy with molecular docking and molecular dynamics simulations to study protein-ligand interactions.
- Investigated ligand binding to human serum albumin and haemoglobin and assessed associated conformational changes.
- Performed molecular modelling and binding free-energy calculations using MM/PBSA and MM/GBSA approaches.
- Evaluated biological activity using H1299 non-small-cell lung cancer cell-based cytotoxicity assays.

EDUCATION

PhD in Chemistry | University of Delhi, India | 2026

Thesis: Design and Exploration of Ionic Liquids as Bioactive Agents: Spectroscopy, Computational and AI Approaches

MSc in Chemistry | Dr. B. R. Ambedkar National Institute of Technology (NIT), Jalandhar, India | 2018

Thesis: Novel Xanthan-Agar Blend and Cadmium Sulphide-Based RSM-CCD Optimized Nanocomposite Hydrogel

BSc (Hons) in Chemistry | University of Delhi, India | 2016

TECHNICAL SKILLS

Experimental: Organic synthesis; UV–Vis; fluorescence spectroscopy; circular dichroism (CD); FRET; FTIR; NMR; HRMS; MTT cell-viability assays.

Computational: Molecular docking; molecular dynamics; protein–ligand interaction analysis; MM/PBSA and MM/GBSA binding free-energy calculations.

Software: GROMACS; Schrödinger Suite; AutoDock; PyMOL; MDAnalysis; Python.

SELECTED PUBLICATIONS

First-author publications and manuscript

- Panchal, S.; Sehrawat, H.; Yadav, N.; Mishra, V.; Sharma, N.; Chandra, R. Anti-Bacterial evaluation of 1,3-Benzodioxole with imidazolium and pyridinium based ionic liquids. Journal of Ionic Liquids (2025), 100130. https://doi.org/10.1016/j.jil.2024.100130
- Panchal, S.; Sehrawat, H.; Sharma, N.; Chandra, R. Biochemical interaction of human hemoglobin with ionic liquids of noscapinoids: Spectroscopic and computational approach. International Journal of Biological Macromolecules 239 (2023), 124227. https://doi.org/10.1016/j.ijbiomac.2023.124227
- Panchal, S.; Mishra, N.; Sehrawat, H.; Sharma, N.; Chandra, R. π–Cation Engineered Noscapine Ionic Liquid Hijacks Serum Albumin via Site I Reprogramming for Enhanced Lung Cancer Therapy. **Manuscript under preparation**.

Co-author publications

- Kritika; Kubba, R.; Kumar, L.; Panchal, S.; Kumar, A.; Roy, I. Phosphorus Corrole–hemoglobin interaction: Exploring binding mechanisms via spectroscopy and molecular simulations studies. Scientific Reports (2026). https://doi.org/10.1038/s41598-026-37177-8
- Sewariya, S.; Mishra, N.; Panchal, S.; Kukreti, S.; Chandra, R. Probing into 9'-Substituted Suzuki-Coupled Noscapine Ionic Liquids as Potent Microtubule Targeting Anticancer Agents with Hemoglobin Affinity. Scientific Reports (2025). https://doi.org/10.1038/s41598-025-94336-z
- Singh, S.; Sewariya, S.; Goel, T.; Panchal, S.; Singh, A.; Kukreti, S.; Tiwari, M. Assessing the Dual Drug 9-hydroxymethyl Noscapine and Telmisartan-loaded Stearic Acid Nanoparticles against H1299 non-small-cell lung cancer and its mechanistic interaction with Bovine Serum Albumin. Materials Advances (2025). https://doi.org/10.1039/D4MA00958D
- Sehrawat, H.; Kumar, N.; Panchal, S.; Kumar, L.; Chandra, R. Imperative persistent interaction analysis of anticancer noscapine-ionic liquid with calf thymus DNA. International Journal of Biological Macromolecules 220 (2022), 415–425. https://doi.org/10.1016/j.ijbiomac.2022.08.106
- Rathee, G.; Kohli, S.; Panchal, S.; Singh, N.; Awasthi, A.; Singh, S.; Singh, A.; Hooda, S.; Chandra, R. Fabrication of a Gold-Supported NiAlTi-Layered Double Hydroxide Nanocatalyst for Organic

Transformations. ACS Omega 5 (2020), 23967–23974. https://doi.org/10.1021/acsomega.0c03250

- Sharma, A. K.; Priya; Kaith, B. S.; Panchal, S.; Bhatia, J. K.; Bajaj, S.; Tanwar, V.; Sharma, N. Response surface methodology directed synthesis of luminescent nanocomposite hydrogel for trapping anionic dyes. Journal of Environmental Management 231 (2019), 380–390. https://doi.org/10.1016/j.jenvman.2018.10.038
- Sharma, A. K.; Priya; Kaith, B. S.; Bajaj, S.; Bhatia, J. K.; Panchal, S.; Sharma, N.; Tanwar, V. Efficient capture of eosin yellow and crystal violet with high performance xanthan-acacia hybrid super-adsorbent optimized using response surface methodology. Colloids and Surfaces B: Biointerfaces 175 (2019), 314–323. https://doi.org/10.1016/j.colsurfb.2018.12.017
- Sharma, A. K.; Priya; Kaith, B. S.; Sharma, N.; Bhatia, J. K.; Tanwar, V.; Panchal, S.; Bajaj, S. Selective removal of cationic dyes using response surface methodology optimized gum acacia-sodium alginate blended superadsorbent. International Journal of Biological Macromolecules 124 (2019), 331–345. https://doi.org/10.1016/j.ijbiomac.2018.11.213
- Priya; Sharma, A. K.; Kaith, B. S.; Tanwar, V.; Bhatia, J. K.; Sharma, N.; Bajaj, S.; Panchal, S.; Chandra, R. RSM–CCD optimized sodium alginate/gelatin based ZnS-nanocomposite hydrogel for the effective removal of biebrich scarlet and crystal violet dyes. International Journal of Biological Macromolecules 129 (2019), 214–226. https://doi.org/10.1016/j.ijbiomac.2019.02.034

CONFERENCE PRESENTATIONS

- Panchal, S.; Sharma, N. (2024). “Biochemical interaction of human hemoglobin with ionic liquids of noscapinoids: Spectroscopic and computational approach.” International Conference on Crossroads of Chemistry, Biology and Atmospheric Environment, University of Delhi, India. Poster presentation.
- Panchal, S.; Sharma, N. (2022). “Synthesis and Interaction of 9-bromo noscapine with human hemoglobin: Spectroscopic and Computational studies.” International Conference on Role of Molecular Chemistry in Drug Design and Development, Amity University, Noida, India. Poster presentation.
- Panchal, S.; Sharma, N. (2022). “Noscapine alkaloid based Ionic liquids as chemotherapeutic agent.” International Conference on Recent Advances in Nano Medical Sciences (RANMS-2022), University of Delhi, India. Poster presentation.

FELLOWSHIPS AND AWARDS

- Joint UGC–CSIR Senior Research Fellowship (Government of India), 2025–2026
- Joint UGC–CSIR Junior Research Fellowship (Government of India), 2023–2025
- University Fellowship (UGC), 2020–2023
- 42nd rank (All India), CSIR–UGC NET, Lectureship, 2023
- 9th rank (All India), DUET PhD Entrance, 2019
- Qualified IIT-JAM, DUET-MSc, and JEE-Mains

RESEARCH INTERESTS

Integrative biomolecular modelling; protein-ligand interactions; protein conformational dynamics; molecular biophysics; structural biology; chemical biology; structure-based drug discovery; allosteric regulation.

REFERENCES

- Prof. Ramesh Chandra, FRSC — Senior Professor, Department of Chemistry, University of Delhi; Founder Director, Dr. B. R. Ambedkar Center for Biomedical Research. acbrdu@hotmail.com
- Prof. Prashant Singh — Professor, Department of Chemistry, Atma Ram Sanatan Dharma College, University of Delhi. psingh@arsd.du.ac.in
- Prof. Neera Sharma — Professor, Department of Chemistry, Hindu College, University of Delhi. neera01@gmail.com